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Unit converter

A few things to add:

1. Add `toppne` to the frame using `frm.add(toppne, BorderLayout.NORTH);`
2. Create another `JPanel` for the two rows in the center.
3. Use `GridLayout(2, 2)` to arrange the two `JTextField`s and two `JComboBox`es.
4. Add an `ActionListener` so the unit options change when you select Length, Weight, or Temperature.

Here's the complete code based on yours:

Java

```java
import javax.swing.JComboBox;
import javax.swing.JFrame;
import javax.swing.JPanel;
import javax.swing.JTextField;
import java.awt.*;
import java.awt.event.*;

public class HW2 extends JFrame {
    static HW2 frm = new HW2();
    static JPanel toppne = new JPanel();
    static JPanel centerpne = new JPanel();

    public static void main(String args[]) {
        frm.setTitle("Unit Converter");
        frm.setLayout(new BorderLayout());
        frm.setSize(480, 280);
        frm.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

        // NORTH: Type selection
        String[] type = {"Length", "Weight", "Temperature"};
        JComboBox<String> typeBox = new JComboBox<>(type);

        toppne.add(typeBox);
        frm.add(toppne, BorderLayout.NORTH);
```

Java

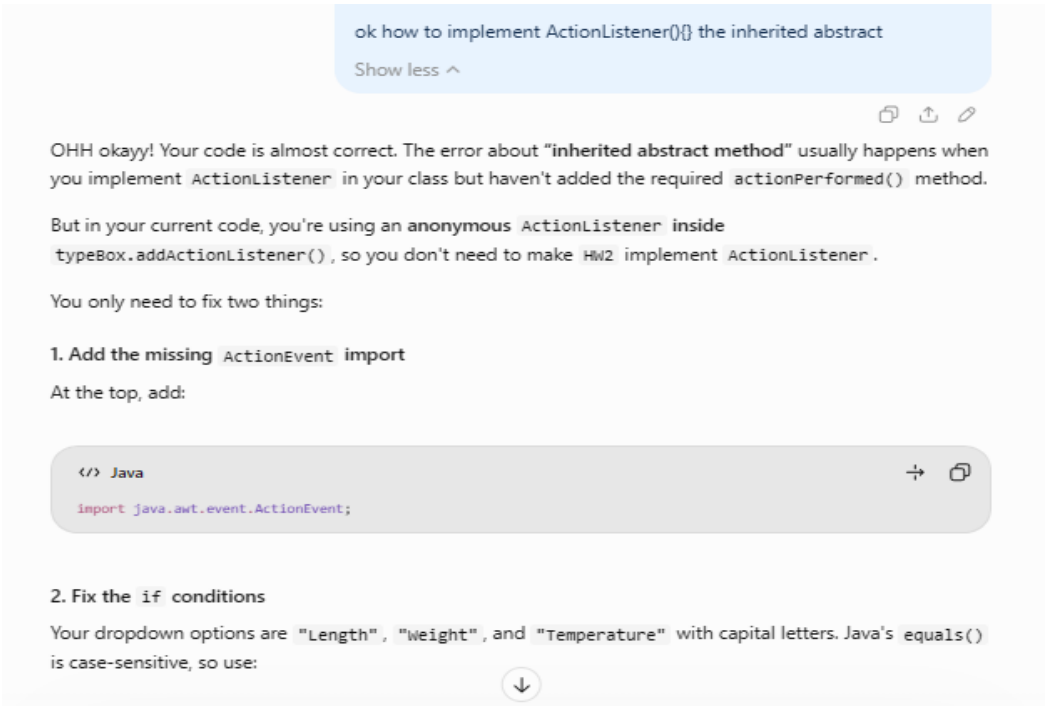
```java
        double answer = value;

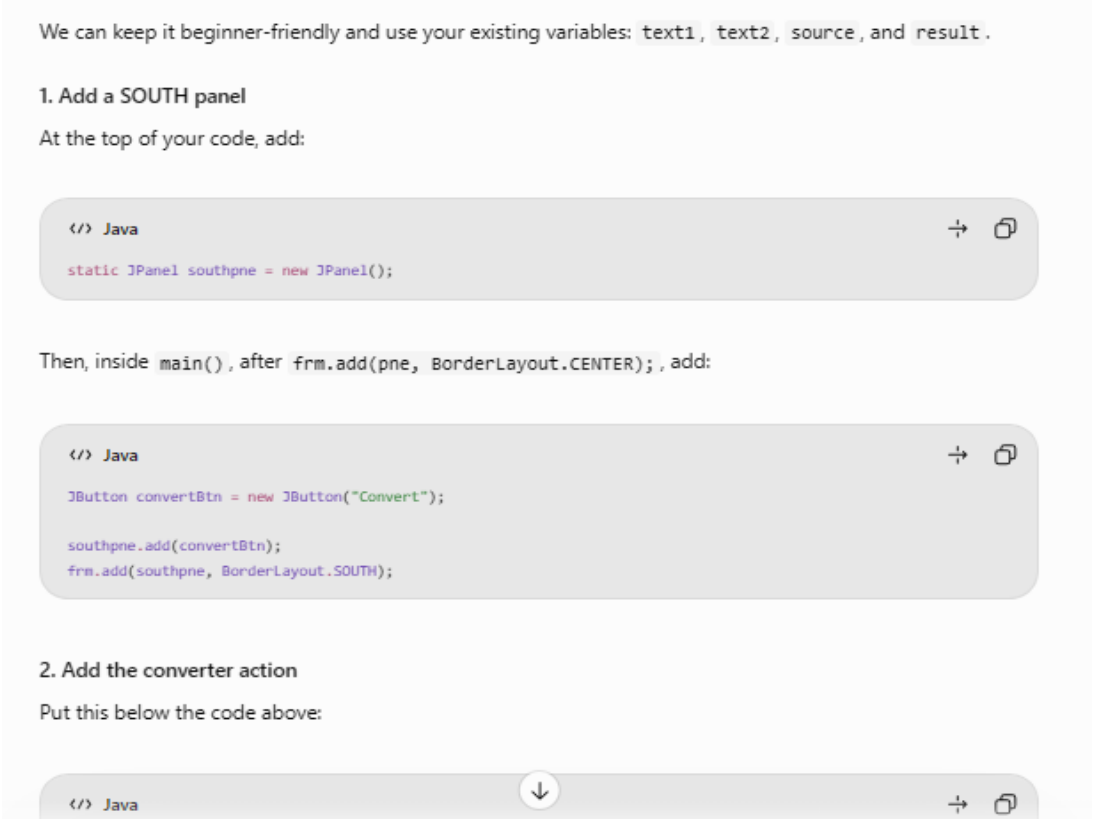
        String from = (String) source.getSelectedItem();
        String to = (String) result.getSelectedItem();
        String type = (String) typeBox.getSelectedItem();

        // Length
        if (type.equals("Length")) {
            if (from.equals("Meters")) {
                answer = value;
            } else if (from.equals("Centimeters")) {
                answer = value / 100;
            } else if (from.equals("Inches")) {
                answer = value * 0.0254;
            } else if (from.equals("Feet")) {
                answer = value * 0.3048;
            }

            if (to.equals("Centimeters")) {
                answer = answer * 100;
            } else if (to.equals("Inches")) {
                answer = answer / 0.0254;
            } else if (to.equals("Feet")) {
                answer = answer / 0.3048;
            }

        // Weight
        } else if (type.equals("Weight")) {
            if (from.equals("Kilograms")) {
                answer = value;
            } else if (from.equals("Grams")) {
                answer = value / 1000;
            } else if (from.equals("Pounds")) {
                answer = value * 0.453592;
            } else {
                answer = value * 0.0283495;
            }
```





Reflection

一開始 AI 給的程式碼對我來說有點複雜，所以我請 AI 改成比較簡單的寫法。我也遇到一些問題，例如 ActionListener 的用法，還有比較 String 時要用 .equals()，不能直接用 ==。後來我修改程式，讓使用者選擇長度、重量或溫度時，單位選單可以跟

著改變。換算的部分，我學到可以先把輸入的數值轉成基本單位，再轉成自己選的目標單位。這樣比較容易理解，也不用寫太多不同的換算公式。透過這次作業，我也更了解 JComboBox、JTextField 和 ActionListener 的用法。